Supplementary Information

Pareto-guided Monte Carlo tree search for analysing multi-objective alternatives in antimalarial molecular design

This supplementary information provides extended scoring definitions, additional candidate-level results, molecular-diversity analysis, scalar benchmark details and algorithmic ablations. The scalar benchmark uses the v12 public-activity reward; the candidate-level Pareto analysis uses the defined secondary pre-activity vector of MPO, SYBA, RRS-informed chemotype similarity and PNS-informed docking proxy.

S1 Extended methods and scoring definitions

Molecular generation. Molecules were constructed from a curated fragment vocabulary using a scaffold-aware policy and a valence filter. The medium vocabulary was used for the main four-method benchmark. Each episode began from benzene and allowed at most 10 fragment additions. The retained MCTS configuration used $c_{\mathrm{PUCT}} = 5.0$, virtual loss $\nu = 0.01$, progressive widening $k = 1.0$ and $\alpha = 0.5$, and policy temperature $T = 0.8$. The policy combined fragment frequencies, scaffold similarity and reactivity compatibility.

Pre-activity objective vector. For the secondary Pareto analysis, each candidate was represented by

$$\mathbf{f}(s) = \big(f_{\mathrm{MPO}}(s), f_{\mathrm{SYBA}}(s), f_{\mathrm{RRS}}(s), f_{\mathrm{PNS}}(s)\big).$$

MPO, SYBA, the RRS-informed chemotype proxy and the PNS-informed docking proxy were maximised; SA was constant at 3.0 and was therefore not used as an independent dominance dimension. The tree-selection value was a scalar sum of the varying min–max-normalised components, while the complete vector was retained for front maintenance and analysis.

The RRS-informed proxy was calculated from Morgan/Tanimoto similarity to the P2 resistance-resilience reference chemotypes. The PNS-informed proxy was the normalised aggregate of the three available Tartarus docking columns (`score_1syh`, `score_6y2f` and `score_4lde`). These quantities are computational proxy dimensions: they are not direct WT/mutant binding measurements and do not constitute a four-target biological assay.

The search-time Pareto records contained a constant SYBA field. SYBA was evaluated separately with the specified classifier for the displayed front; MPO, SA, RRS and PNS were unchanged. This preserves the candidate discovery record while allowing accessibility to be displayed as a non-constant secondary dimension. The resulting front should consequently be interpreted as a secondary objective analysis, not as evidence that informative SYBA values guided the search.

Scalar benchmark. The v12 benchmark used a separate scalar reward with weights $w_{\mathrm{MPO}} = 0.36$, $w_{\mathrm{docking}} = 0.315$, $w_{\mathrm{SYBA}} = 0.135$, $w_{\mathrm{SA}} = 0.09$ and $w_{\mathrm{activity}} = 0.10$. RRS and PNS were not included in this benchmark. Public-activity proximity was the maximum Morgan-2 Tanimoto to experimentally active compounds in the independent ChEMBL $\mathrm{IC}_{50}/\mathrm{EC}_{50}$ panel (22 447 molecules, including 19 321 active entries).

S2 Pareto-front candidates and domain-proxy profiles

Using the separate secondary pre-activity objective set, the merged front contained four non-dominated candidates from 20 independent seeds. Their scores and seed-level values are shown in **??**. The hypervolume was 1.236 6 under min–max normalisation and reference point $\mathbf{r} = 1.1$. Because this value depends on the normalisation convention, it is used for comparison within the specified protocol rather than as an absolute measure of chemical quality.

The front spans a potency–accessibility trade-off. The third candidate has the largest MPO but the lowest SYBA, whereas the first, second and fourth candidates have SYBA values above 0.99. Three candidates have PNS equal to 1.0, the maximum value of the normalised three-column docking proxy. The RRS-informed

Table S1: Pre-activity Pareto-front candidates and their objective values. SYBA was evaluated separately before the front was constructed; the other displayed values are unchanged from the candidate records.

Candidate (SMILES)	MPO	SYBA	SA	RRS	PNS	Seed
CNC(=O)c1ccc(OC(C)C(C)(C)C)c(OC)c1	0.910	1.000	3.0	0.211	1.0	13
COc1c(C)cc(-c2ccc(S(N)(=O)=O)cc2)c c1C	0.945	1.000	3.0	0.196	1.0	18
CC1CCC(n2cccn2)NC1(c1ccccc1)C1O CCO1	0.946	0.021	3.0	0.141	0.0	5
COc1cc(CO)c(O)c(C)c1C	0.729	0.994	3.0	0.223	1.0	6

proxy ranges from 0.141 to 0.223. These values are useful for candidate stratification; they do not establish polypharmacological binding or resistance resilience experimentally.

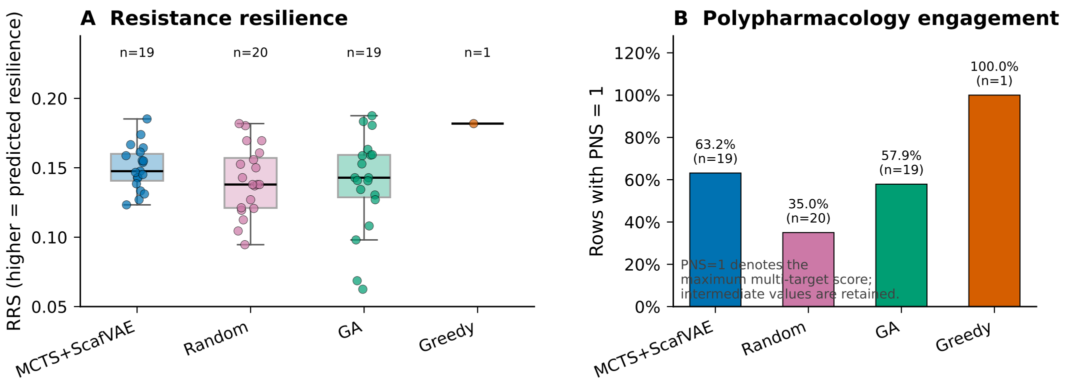

Figure S1: RRS-informed chemotype-similarity and PNS-informed docking-proxy profiles for recorded best-in-seed molecules. Points show individual observations and box summaries. The PNS panel reports the fraction of valid rows with PNS equal to 1.0; intermediate values are retained in the underlying data.

S3 Molecular diversity

For each generation method, the best molecule from each of the 20 benchmark seeds was retained. ECFP4 Tanimoto dissimilarity and Bemis–Murcko scaffold statistics were calculated with RDKit. The three stochastic methods show non-zero pairwise dissimilarity under the present fragment vocabulary, whereas greedy search returns one repeated local optimum. These statistics characterise the sampled candidate sets.

Table S2: Molecular diversity of the four generation methods. Pairwise dissimilarity is the mean of $1 - \text{Tanimoto}$ over the 190 unique pairs per method; scaffolds are unique Bemis–Murcko skeletons.

Method	n	Mean pairwise dissimilarity	Unique scaffolds	Scaffold fraction
MCTS+ScafVAE	20	0.773 2	19	0.950
Random	20	0.804 6	20	1.000
Greedy	20	0.000 0	1	0.050
GA	20	0.776 1	12	0.600

Random search has full scaffold coverage, whereas MCTS and the genetic algorithm yield 19 and 12 unique scaffolds, respectively. These statistics describe the sampled molecule sets and are not intended as a claim that one method explores all of chemical space more effectively.

S4 Extended scalar benchmark

The scalar benchmark comprised 20 independent seeds per method and a fixed budget of 1 000 search iterations. The values in ?? are the mean, standard deviation, observed range and mean wall-clock time of the best scalar reward per seed.

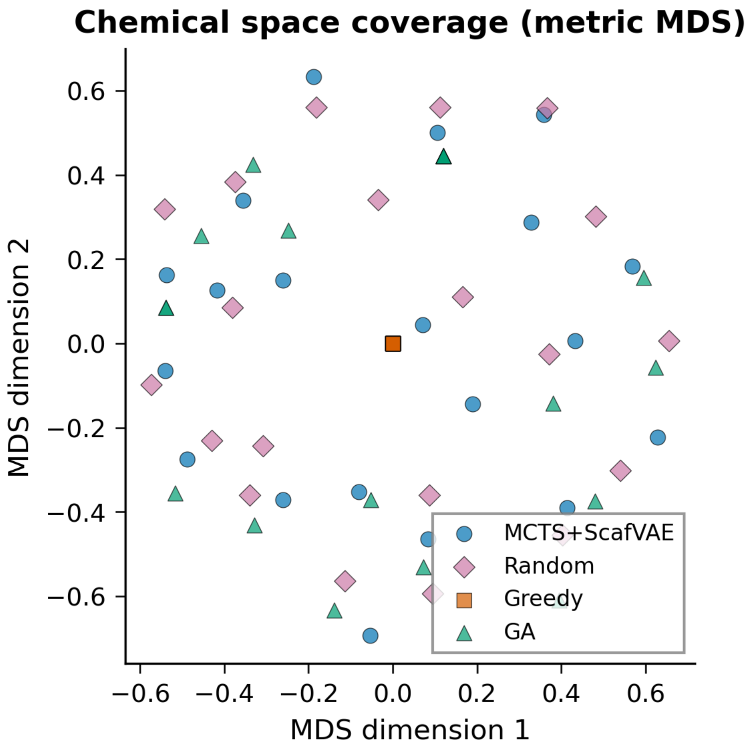

Figure S2: Metric multidimensional-scaling projection of the recorded best-in-seed molecule sets using ECFP4 Tanimoto dissimilarity. Each method contributes 20 molecules. Greedy search collapses to one point because it returns the same local optimum for every seed.

Table S3: Public-activity-augmented scalar benchmark across 20 seeds. Reward values are reported with the screened MCTS configuration; wall-clock time is the mean per seed.

Method	Mean reward	SD	Minimum	Maximum	Mean time (s)
Random	0.672 4	0.005 6	0.661 6	0.685 6	50.4
MCTS+ScafVAE	0.664 9	0.006 8	0.655 4	0.677 9	92.1
Greedy	0.427 8	0.000 0	0.427 8	0.427 8	1.9
Genetic algorithm	0.645 3	0.012 4	0.617 8	0.671 5	2.0

Random search has the highest mean scalar reward, followed by MCTS, the genetic algorithm and greedy search. The scalar benchmark measures search efficiency; the Pareto analysis measures which candidate-level trade-offs remain available. These are complementary outcomes rather than a single ranking.

S5 Algorithmic ablations

A factorial ablation evaluated the chemistry-informed policy, Pareto-front maintenance, PUCT exploration, rollout temperature and fragment vocabulary. Each factorial cell used five replicates. Because this experiment uses a distinct reward configuration, its factor effects are interpreted within the ablation only. The marginal factor means are summarised in **??**. The largest observed effects were associated with the ScafVAE policy ($\Delta = +0.148$), Pareto-front maintenance ($\Delta = +0.108$) and the larger fragment vocabulary ($\Delta = +0.079$). Changing c_{PUCT} from 2.0 to 1.0 improved the reward by $\Delta = +0.055$, whereas temperature had a smaller effect.

Table S4: Marginal effects in the factorial ablation. Each factor level aggregates 80 observations from the 32-cell design with five replicates per cell. The difference is calculated as the mean for level B minus the mean for level A; it is an within-ablation effect and is not comparable with the v12 scalar benchmark.

Factor	Level A	Level B	n per level	Δ
ScafVAE	False: 0.630	True: 0.778	80	0.148
Pareto front	False: 0.650	True: 0.758	80	0.108
c_{PUCT}	2.0: 0.677	1.0: 0.731	80	0.055
Temperature	1.5: 0.698	0.5: 0.710	80	0.012
Vocabulary	Small: 0.665	Large: 0.743	80	0.079

A separate vocabulary comparison across 10 seeds per condition is shown in **??**. The aromatic-only vocabulary performed below the full, medium and minimal vocabularies; the medium and minimal conditions differed modestly ($p_{\mathrm{adj}} = 0.005\,2$), whereas the full and medium conditions did not differ significantly.

Table S5: Fragment-vocabulary ablation. Values are mean reward ± SD over 10 seeds per condition. Tukey HSD was used for pairwise comparisons.

Vocabulary	n	Mean reward	SD
All	10	0.624 0	0.005 6
Aromatic-only	10	0.549 7	0.003 0
Medium	10	0.628 3	0.004 8
Minimal	10	0.618 3	0.009 7

The selection mechanism was also compared with a ParetoPUCT-style variant that restricted each node to non-dominated children before applying the PUCT score. At equal budget (700 iterations and 5 seeds), the merged hypervolumes were 15.56 for scalar-proxy selection and 15.49 for Pareto-restricted selection. Within this sensitivity experiment, the two selection rules produced similar merged hypervolumes; neither rule showed a measurable advantage at the tested budget.

S6 Data and code availability

Molecule sets, benchmark records, Pareto scores, diversity statistics and ablation summaries are provided with the article's open computational materials. The accompanying scripts regenerate the reported analyses.